

Thapar Institute of Engineering and Technology, Patiala

Mechanical Engineering Department

TUTORIAL SHEET-1

B. E. (MT/RAI) (Sem-5)
(2026-27)

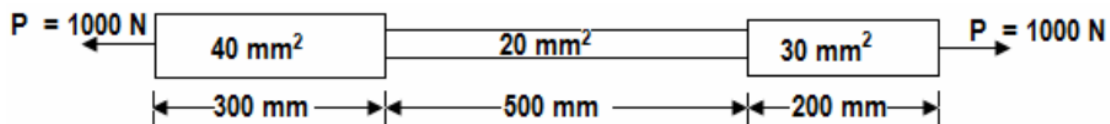
Course Code: UMT403
Course: Stress analysis and Mechanical Design

Q1. A Mild Steel wire 5 mm in diameter and 1 m long. If the wire is subjected to an axial tensile load 10 kN find its extension of the rod. ($E = 200 \text{ GPa}$).

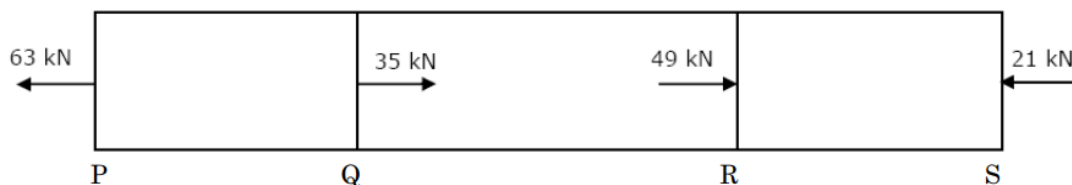
(Ans. 2.55 mm)

Q2. The composite rod shown below is 1000 mm long, its two ends are 40 mm^2 and 30 mm^2 in area and length are 300 mm and 200 mm respectively. The middle portion of the rod is 20 mm^2 in area and 500 mm long. If the rod is subjected to an axial tensile load of 1000 N, find its total elongation. ($E = 200 \text{ GPa}$)

(Ans. 0.196 mm)

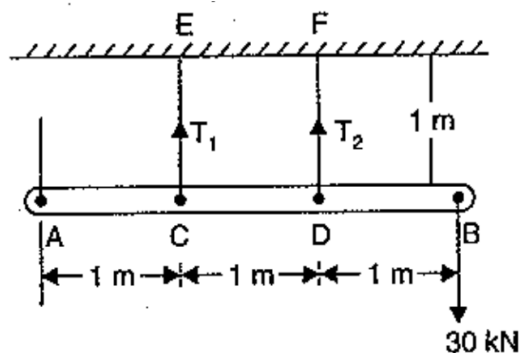


Q3. A bar having a cross-sectional area of 700 mm^2 is subjected to axial loads at the positions indicated. Calculate the stress in segment QR. (Ans. 40 MPa)



Q4. A rigid bar ABCD shown below is hinged at 'A' and supported in a horizontal position by two identical steel wires as shown in figure below. A vertical load of 30 kN is applied at 'B'. Find the tensions T_1 , T_2 induced in both the wires by the vertical load and hence stresses, if each wire is of 20 mm diameter.

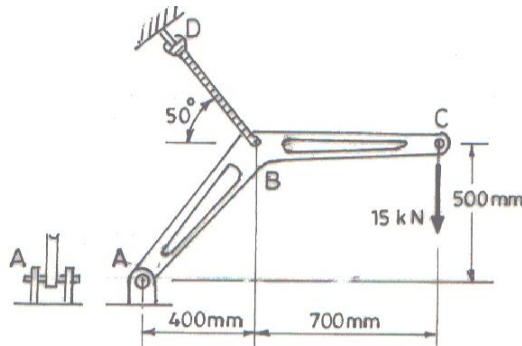
(Ans. $T_1 = 18 \text{ kN}$, and $T_2 = 36 \text{ kN}$, $\sigma_1 = 57.3 \text{ N/mm}^2$, $\sigma_2 = 114.6 \text{ N/mm}^2$)



Q5. A block $100 \text{ mm} \times 100 \text{ mm}$ base and 10 mm height. When we apply a tangential force 10 kN to the upper edge it is displaced 1 mm relative to lower face. Find the direct shear stress and shear strain in the element.

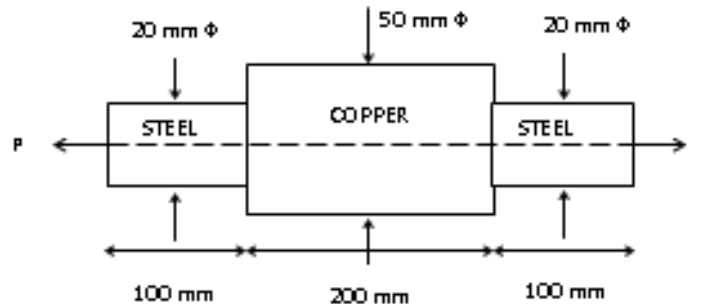
(Ans. 1 MPa, 0.1 dimensionless)

- Q6. Determine the tensile stress in cable BD and shearing stress in the pin at A. The diameter of cable and pin are 20 mm and 10 mm, respectively. Draw the free body diagram of ABC. (Answers: $\sigma_{BD}=83.65 \text{ MPa}$, $\tau_{pin}=112.37 \text{ MPa}$)



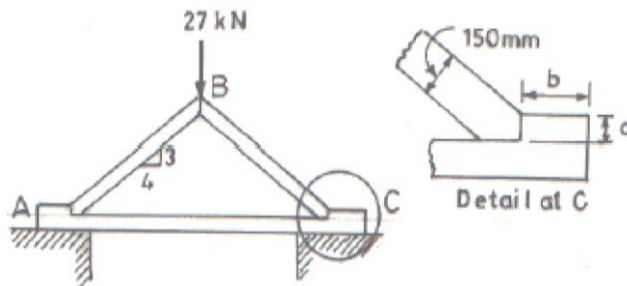
- Q7. A composite bar formed by joining two concentric steel rods of 20 mm diameter and 100 mm length to each end of a copper rod of 50 mm diameter and 200 mm length. When an axial force is applied to the bar, its total extension is found to be 0.2 mm. Determine the axial stresses in the copper and steel rods. Take $E_{steel} = 210 \text{ GPa}$ and $E_{copper} = 120 \text{ GPa}$.

(Answers: $\sigma_{st}=164.06 \text{ MPa}$, $\sigma_{cu}=26.25 \text{ MPa}$)

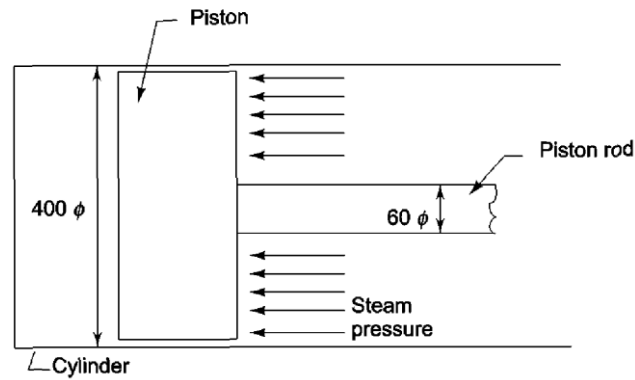


- Q8. For the timber truss shown below, the allowable stresses in direct bearing and shear are 6 N/mm^2 and 1.2 N/mm^2 , respectively. Compute the proper dimension **a** and **b** for the notched joints at the ends of the horizontal member. The size of the timber members is $150 \text{ mm} \times 150 \text{ mm}$. Assume joint B behaves like a pinned connection.

(Answers: $a=20 \text{ mm}$, $b=100 \text{ mm}$)

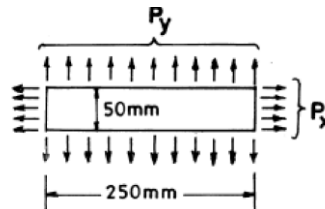


- Q9. The piston of a steam engine is 60 mm diameter and operates in a cylinder of diameter 400 mm. The piston rod is 1 m long. What is the maximum pressure that can be allowed in the cylinder, if the stress in the rod is limited to 100 N/mm^2 ? What will be the change in the length of the piston at this pressure? $E = 200 \text{ GPa}$. (Ans: 2.30 N/mm^2 , 0.5 mm)



Q10. A piece of 50 x 250 x 10 mm steel plate is subjected to uniformly distributed stresses along its edges as shown in Fig. ($E=200$ GPa and $\nu=0.25$).

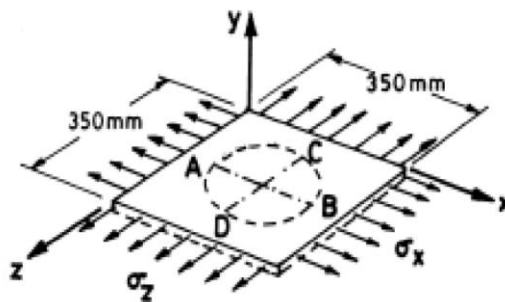
- If $P_x = 100$ kN and $P_y = 200$ kN then determine the change in thickness of plate.
- Determine the P_x alone which will cause the same change in thickness as in (a).



(Answers: $\delta_t = 0.0035$ mm, $P_x = 140$ kN)

Q11. A circle of diameter 200 mm, is scribed on an unstressed 18 mm thick aluminium plate as shown in Fig. Forces acting in the plane of the plate later cause normal stresses $\sigma_x = 85$ MPa and $\sigma_z = 150$ MPa. ($E=70$ GPa and $\nu=1/3$). Determine the

- change in the length of diameter AB and CD
- change in the thickness of plate



(Answers: $\delta L_{(AB)} = 0.1$ mm; $\delta L_{(CD)} = 0.348$ mm; $\delta t = -0.020$ mm)

Q12. A 2 m long solid circular rod of diameter 12 mm is subjected to an axial tensile force. The shear modulus of the material is 80 GPa and Poisson's ratio is 0.25. The rod elongates by 10 mm due to the axial tensile force. Determine the final diameter of the rod.

(Answer: 11.985 mm)

Q13: A 500 mm long, 16 mm diameter rod is observed to increase in the length by 300 μ m, and to decrease in the diameter by 2.4 μ m when subjected to an axial load of 25 kN. Determine the modulus of elasticity, Poisson's ratio and shear modulus.

(Answers: 207.23 GPa; 0.25; 82.9 GPa)